

# WS8856FLS

## Wi-SUN Module

### Instruction Manual

- Support wide frequency band 861-930MHz
- Support 1000 node-shaped networks
- Support high air data rate up to 300kbps
- Support WI-SUN FAN 1.0 protocol stack
- Support multi-level power consumption 20mW, 10mW, 1mW adjustable
- Support multi-channel frequency hopping transmission
- Support bidirectional communication between repeaters and terminals to establish a mesh network
- Support multi-hop network between repeaters (relay function)
- Support serial port command parameter configuration

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# 1 Instruction format introduction

Instruction format:

Command Name [Parameter 1] [Parameter 2].. [Parameter n] < Carriage Return > < Line Feed >

Use a space to separate ( "). The instruction name is composed of lowercase English letters, and the parameters are composed of numbers or strings. If there is no parameter after the instruction name, the parameter query is executed. If there is a parameter after the instruction name, the parameter configuration is executed. The instruction ends with "< CR > < LF >" (i.e. "\ R \ n").

Example: udps 2001: db8: 8C 12345678 < CR > < LF >

A udps represent transmitted UDP data and a 12345678 represent transmitted data.

## 2 Common instructions

### 2.1 Save parameter saving

Command overview: This command saves the parameters to flash.

Instruction content:

| Send | Respond |
|------|---------|
| save | ok      |

Command parameter: none

### 2.2 Reset of clear parameter

Command overview: This command restores the parameters to the factory settings.

Instruction content:

| Send  | Respond |
|-------|---------|
| clear | ok      |

Command parameter: none

### 2.3 Help Query Command Function

Command overview: This command queries the AT command set supported by the current module.

Instruction content:

| Send | Respond                     |
|------|-----------------------------|
| help | Instruction set description |

Command parameter: none

## 2.4 Reset module restart

Command overview: This command performs a soft restart of the module.

Instruction content:

| Send        | Respond                  |
|-------------|--------------------------|
| reset       | OK (restart immediately) |
| reset [sec] | OK (delayed restart)     |

Command parameters:

|       |                                                                   |
|-------|-------------------------------------------------------------------|
| [sec] | Restart after a delay of how many seconds, parameter range 1-3600 |
|-------|-------------------------------------------------------------------|

## 2.5 Version Query version information

Command overview: This command queries the software version and hardware version of the module

Instruction content:

| Send    | Respond                                                              |
|---------|----------------------------------------------------------------------|
| version | Part Number: <PN><br>Software Version:<SV><br>Hardware Version: <HV> |

Command parameter: none

## 2.6 Save and restart the svrst parameter

Command overview: This command saves the parameters to flash and then executes the module restart command.

Instruction content:

| Send        | Respond                                          |
|-------------|--------------------------------------------------|
| svrst       | OK (restart immediately after saving parameters) |
| svrst [sec] | OK (delayed restart after saving parameters)     |

Command parameters:

|       |                                                                   |
|-------|-------------------------------------------------------------------|
| [sec] | Restart after a delay of how many seconds, parameter range 1-3600 |
|-------|-------------------------------------------------------------------|

## 2.7 Clrst parameter reset and restart

Command overview: This command restores the factory settings, and then executes the module restart command to take effect.

Instruction content:

| Send  | Respond                                        |
|-------|------------------------------------------------|
| clrst | OK (restart immediately after parameter reset) |

clrst [sec]

OK (delayed restart after parameter reset)

Command parameters:

|       |                                                                   |
|-------|-------------------------------------------------------------------|
| [sec] | Restart after a delay of how many seconds, parameter range 1-3600 |
|-------|-------------------------------------------------------------------|

## 2.8 Param query wisun basic parameters

Command overview: Query the basic parameters of wi-sun network.

Instruction content:

| Send  | Respond                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| param | [Network role]: <role><br>[Network state]: <state><br>[Network name]: <name><br>[Mac address]: <mac><br>[Pan id]: <id><br>[Ipv6address]: <ip>(local) <ip>(global)<br>[Tx power]: <power><br>[CCA threshold]: <cca><br>[Wi-sun PHY]: <domain> <rate> <class><br><base_channel> <channel_space> <channel_number><br>[UDP options]: <udp_port> <broadcast> <multicast> <disp_source> |

Command parameters:

|                 |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [Network role]  | Wi-sun network roles: 1-border router, 2-router.                                                                                                                                                                                                                                                                                                                                                              |
| [Network state] | Network connection status 1: Ready to start connection. Select the network to connect to.<br>Network connection state 2: During network authentication. Network connection status 3:<br>FAN configuration. Network Connection Status 4: Routing Configuration. Network<br>connection status 5: The network is in normal operation, and communication can be carried<br>out after network access is completed. |
| [Network name]  | Wi-sun network name. Only the same network name can access the network. 1 to 32 ASCLL characters.                                                                                                                                                                                                                                                                                                             |
| [Mac address]   | 64-bit MAC address of the device.                                                                                                                                                                                                                                                                                                                                                                             |
| [Pan id]        | Wi-sun network PANID, parameter range 0x0001-0xffff.                                                                                                                                                                                                                                                                                                                                                          |
| [Ipv6address]   | IPv6 addresses, including link-local and global unicast addresses.                                                                                                                                                                                                                                                                                                                                            |
| [Tx power]      | Module RF transmit power, ranging from -20 dBm to 14 dBm.                                                                                                                                                                                                                                                                                                                                                     |
| [CCA threshold] | Idle channel evaluation threshold, the default is -60dBm, and the parameter range is -60dBm to -100dBm.                                                                                                                                                                                                                                                                                                       |
| [Wi-sun PHY]    | PHY information of wi-sun, including wi-sun area domain, air rate, frequency channel information, etc.                                                                                                                                                                                                                                                                                                        |

[UDP options]

Option configuration of UDP communication, including port, multicast address, broadcast address, received data packet and display source address

## 2.9 Role query wisun network roles

Instruction overview: Query the roles that the wi-sun network is running, including Border and router.

Instruction content:

| Send | Respond         |
|------|-----------------|
| role | net role:[role] |

Command parameters:

[role]                      Wi-sun network roles: 1-border router, 2-router.

## 2.10 Netname Wi-SUN network name configuration and query

Command overview: This command sets the network name of Wi-SUN. Only the same network name can join the network. The default network name of the module is Wi-SUN Network. For example, border creates a network with a network name of MyNetwork. The network name of the router node is configured as MyNetwork before it can join the network.

Instruction content:

| Send           | Respond             |
|----------------|---------------------|
| netname [name] | ok                  |
| netname        | Network name:[name] |

Command parameters:

[name]                      Indicates the name of the Wi-Sun network to be configured. The argument range is 1 to 32 ASCLL characters.

## 2.11 Pan Wi-SUN Network PANID Configuration and Query

Command overview: configure the PAN ID of the wisun network and allow the Border router to configure the PANID. The router node is fixed as 0xffff and can be added to the BR of any PANID.

Instruction content:

| Send     | Respond    |
|----------|------------|
| pan [id] | ok         |
| pan      | panid:[id] |

Command parameters:

[id]                      Indicates wi-sun network PANID, parameter configuration range

0x0001-0xffff。

## 2.12 IP IPV6 address query

Command overview: This command can query the local address and global address of IPv6.

Instruction content:

| Send | Respond                                         |
|------|-------------------------------------------------|
| ip   | Ipv6 address:[address](local) [address](global) |

Command parameters:

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| [address] | 128-bit IPV6 address, including link-local and global unicast addr |
|-----------|--------------------------------------------------------------------|

## 2.13 Mac module MAC address query

Command overview: Query the MAC address. If necessary, it is allowed to configure the MAC address. It must be noted that the MAC address cannot be repeated. If the MAC address is repeated, it cannot access the network for communication. The MAC address is related to the IP address. The last 64 bits of the IP address are the same as the MAC address. Therefore, modifying the MAC address can also achieve the function of modifying the IP address.

Instruction content:

| Send          | Respond       |
|---------------|---------------|
| mac           | mac [address] |
| mac [address] | ok            |

Command parameters:

|           |                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------|
| [address] | 64-bit MAC address. If you want to configure the MAC address, ensure that the MAC address cannot be duplicated. |
|-----------|-----------------------------------------------------------------------------------------------------------------|

## 2.14 UDP opts UDP Configuration Option

Command overview: (1) set the UDP port number, (2) set the UDP broadcast address, (3) set the UDP multicast address, (4) set whether to display the source IP address of the received data.

Instruction content:

| Send                 | Respond                                             |
|----------------------|-----------------------------------------------------|
| udpopts [opt] [para] | ok                                                  |
| udpopts              | udp_port=[port]<br>broadcast=[ip]<br>multicast=[ip] |

disp\_source=[on/off]

Command parameters:

| [opt]       | [para]                                                                     |
|-------------|----------------------------------------------------------------------------|
| udp_port    | Port number for receiving and sending UDP data.                            |
| broadcast   | The IP address of the UDP broadcast.                                       |
| multicast   | The IP address of the UDP multicast.                                       |
| disp_source | Whether the source IP address is displayed in the packets received by UDP. |

## 2.15 Udpes UDP data transmission

Command overview: This command uses UDP protocol to send data, which can realize on-demand, multicast and broadcast.

Instruction content:

| Send               | Respond |
|--------------------|---------|
| udps [addr] [data] | ok      |

Command parameters:

|        |                                                                                                                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [addr] | Ipv6 address of the target node                                                                                                                                                                                                                                        |
| [data] | The sending data is an ASCLL character string with a maximum length of 1480 bytes. The data in the form of HEX is sent and received by transparent transmission. It can also be simulated by an ASCLL character string. For example, fad348 represents 0xfa 0xd3 0x48. |

## 2.16 Fstat network connection status query

Command overview: This command queries the current connection status of the network.

Instruction content:

| Send  | Respond       |
|-------|---------------|
| fstat | fstat [state] |

Command parameters:

|         |                                                                                                                                                                                                                                                                                                                     |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [state] | Network connection status 1, ready to start connection, select the connected network<br>Network connection status 2, during network authentication<br>Network connection status 3, FAN configuration<br>Network Connection Status 4, Routing Configuration<br>Network connection status 5, the network is in normal |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

operation, and communication can be carried out after network access is completed

### 2.17 Chrate sets the over-the-air rate of the channel PHY

Command overview: This command sets the air rate of the channel PHY. If the air rate in the wi-sun specification is different, the number of channels, channel spacing, and starting frequency may be different. After the air rate is modified, the chconfig command can query the channel PHY information of the relevant air rate.

Instruction content:

| Send           | Respond     |
|----------------|-------------|
| chrates        | rate:[rate] |
| chrates [rate] | ok          |

Command parameters:

[rate]                      Air speed 50kbps, 100kbps, 150kbps, 200kbps, 300kbps

### 2.18 Domain setting wi-sun zone configuration

Command overview: This command sets the region configuration of wi-sun. Currently, North America, Japan, Europe, Brazil and other regions can be set.

Instruction content:

| Send        | Respond            |
|-------------|--------------------|
| domain      | region domain:[id] |
| domain [id] | ok                 |

Command parameters:

[id]                      Territory ID, 1: NA (North America), 2: JP (Japan), 3: EU (Europe), 7: BZ (Brazil)

NA supported airspeed range: 50Kbps 100Kbps 150Kbps 200Kbps 300Kbps

Airspeed range supported by JP: 50Kbps 100Kbps 150Kbps 200Kbps 300Kbps

EU supported airspeed range: 50Kbps 100Kbps 150Kbps

Airspeed range supported by BZ: 50Kbps 100Kbps 150Kbps 200Kbps 300Kbps

### 2.19 Chconfig Query Channel PHY Configuration

Command overview: This command queries the current network channel configuration information.

Instruction content:

| Send | Respond |
|------|---------|
|------|---------|

chconfig      domain=[domain]    rate=[rate]      class=[class]  
                  chbase=[ch0]    chspace=[spacing]    chnum=[num]

Command parameters:

|          |                                                      |
|----------|------------------------------------------------------|
| [domain] | Territory ID, 1: NA, 2: JP, 3: EU, 7: BZ             |
| [rate]   | Air speed 50kbps, 100kbps, 150kbps, 200kbps, 300kbps |
| [class]  | Operation level: 1,2,3                               |
| [ch0]    | Center frequency of channel 0                        |
| [space]  | Channel spacing                                      |
| [chnum]  | Total number of channel                              |

## 2.20 + + + Switch to passthrough mode

Command overview: Switch to passthrough mode in AT command mode

Instruction content:

|      |                  |
|------|------------------|
| Send | Respond          |
| +++  | Transparent mode |

Command parameter: none

## 2.21 Saving the at \_ save parameter

Command overview: parameters are saved in the passthrough mode, and the passthrough mode can be saved to flash (power-down save), which can realize data communication in the passthrough mode when the module is powered on.

Instruction content:

|         |         |
|---------|---------|
| Send    | Respond |
| at_save | ok      |

Command parameter: none

## 2.22 At \_ dest configures the target IP address in passthrough mode

Command overview: configure the destination IP address for communication in passthrough mode.

Instruction content:

|              |         |
|--------------|---------|
| Send         | Respond |
| at_dest [ip] | ok      |

Command parameters:

[IP]: destination IP address for transparent data transmission

### 2.23 exit Exit passthrough mode

Command overview: Switch to AT mode in passthrough mode

Instruction content:

| Send | Respond |
|------|---------|
| exit | Ok      |

### 2.24 Ping

Command overview: This command sends ICMP commands to test whether the network is clear.

Instruction content:

| Send              | Respond                                                                                |
|-------------------|----------------------------------------------------------------------------------------|
| ping [ipv6]       | [Len] bytes from [IPv6]: ICMP _ seq = [seq] time = [time] ms (default length 40 bytes) |
| ping [ipv6] [len] | [Len] bytes from [IPv6]: ICMP _ seq = [seq] time = [time] ms (data length can be set)  |

Command parameters:

|        |                                                                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [ipv6] | An IP address                                                                                                                                                |
| [len]  | Specify the data length of the ping command. The data length may not be specified. The default data length is 40 bytes, and the maximum length is 1480 bytes |
| [seq]  | The sequence number of the ICMP packet                                                                                                                       |
| [time] | The time required from sending a request command to receiving a response command.                                                                            |

### 2.25 txpower RF transmit power configuration and query

Command overview: Configure and query the module's RF transmit power.

Instruction content:

| Send             | Respond          |
|------------------|------------------|
| txpower          | tx power:[power] |
| tx power:[power] | ok               |

Command parameters:

|         |                                                           |
|---------|-----------------------------------------------------------|
| [power] | Module RF transmit power, parameter range 14dBm to 20dBm. |
|---------|-----------------------------------------------------------|

### 2.26 CCA clear channel assessment threshold configuration and query

Command overview: Configures and queries the CCA (Clear Channel Assessment) threshold of the module.

Instruction content:

|           |                     |
|-----------|---------------------|
| Send      | Respond             |
| cca       | CCA threshold:[dbm] |
| cca:[dbm] | ok                  |

Command parameters:

[dbm] CCA (Clear Channel Assessment) threshold, parameter range -60dBm to -100dBm.

## 2.27 The connum boundary route obtains the number of nodes currently accessed

Command overview: Border router obtains the number of nodes currently accessing the network. This command router is invalid.

Instruction content:

|        |                        |
|--------|------------------------|
| Send   | Respond                |
| connum | connected number:[num] |

Command parameters:

[num] The number of nodes accessed by the Border router.

## 2.28 The conlist boundary route obtains the list of nodes currently accessed.

Command overview: Border router obtains the node list of the current access network. This command router is invalid.

Instruction content:

|         |                         |
|---------|-------------------------|
| Send    | Respond                 |
| conlist | connected devices list: |
|         | <ipv6_0>                |
|         | <ipv6_1>                |
|         | <ipv6_2>                |
|         | .....                   |
|         | <ipv6_n>                |

Command parameters:

[ipv6\_n] IPV6 address list of nodes accessed by Border router  
The last 64 bits of the node's IPV6 are the same as the MAC address.

## 2.29 Routpath boundary routing obtains the routing path of a node in the network.

Command overview: Border router is the routing path of a node in the network. This command can obtain the network topology. The router of this command is invalid.

Instruction content:

|      |         |
|------|---------|
| Send | Respond |
|------|---------|

routpath [ipv6\_dest]

router path :

<ipv6\_0> ->

<ipv6\_1> ->

<ipv6\_2> ->

.....

<ipv6\_dest>

Command parameters:

[ipv6\_n]

< IPv6 \_ 0 > is the address of the Border router, < IPv6 \_ dest > is the address of the destination device, and the middle address is the address of the node through which the routing path passes.

### 2.30 Neighbor get neighbor table

Command overview: Get the surrounding neighbor devices. The device information includes the MAC address.

Instruction content:

Send

Respond

neighbor

current neighbors number:[num]

<mac\_0>

<mac\_1>

<mac\_2>

.....

<mac\_n>

Command parameters:

[num]

Current number of neighbor

[mac]

MAC address of the neighbor device

## 3 Contact information

| Contact person | Cell phone  | Mailbox                    |
|----------------|-------------|----------------------------|
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